Table 1: Rigidity penalties compared (test set, 28 patients): network trained with the PET losses, IO with the PET terms and one rigidity penalty at the weight chosen on the validation set. Rigidity of bones is measured in three ways: the residual of a rigid fit to each bone (the quantity the rigid-fit penalty minimises), the Staring energy, i.e. the Staring penalty (properness, orthonormality, and affinity) on the whole bone (the quantity the Staring penalty minimises), and the distortion of short (5–15 mm) and long ($\geq$15 mm) within-bone distances (minimised by neither). Mean$\pm$SD over patients. $^{\dagger}/^{\ddagger}$: significantly better/worse than without rigidity (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over the three penalties per column, $p < 0.05$).

	Alignment			Rigidity of bones					Preservation	
Rigidity penalty	CT Dice (%)	Bone Dice (%)	Soft-tissue Dice (%)	Rigid-fit residual (mm)	Staring energy	Distortion short (%)	Distortion long (%)	NDV in bone (ppm)	$\lvert\Delta\text{MTV}\rvert$ (%)	$\lvert\Delta\text{TLG}\rvert$ (%)
No rigidity	77.7±4.8	78.4±6.4	79.9±3.9	2.67±0.52	3.517±2.110	13.41±3.62	4.44±0.83	6.5±4.6	8.2±10.1	10.3±10.2
Staring (×3)	75.7±4.3‡	74.6±5.7‡	79.8±3.9	2.00±0.48†	0.272±0.090†	5.53±1.04†	3.17±0.68†	0.3±1.2†	7.1±8.7	9.2±9.4
Staring + erosion (×100)	77.6±4.6	77.5±6.2‡	80.7±3.9†	2.31±0.47†	1.803±0.650†	8.76±1.04†	3.53±0.63†	16.5±23.6‡	7.9±10.0	9.6±10.2
Per-bone rigid fit (×100)	76.1±4.6‡	74.5±6.2‡	80.8±4.0†	1.39±0.40†	1.419±0.473†	5.01±1.13†	1.83±0.62†	9.6±23.7	8.0±8.9	10.5±9.4